

Shantharam P

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SUMMARY

Computer Science & Engineering student with strong foundations in backend development, networking, distributed systems, and problem solving. Experienced in building scalable REST APIs, IoT-enabled backend systems, OTA firmware frameworks, and automation-driven workflows. Skilled in Python, C/C++, Django, SQL, and TCP/IP fundamentals, with hands-on experience in embedded systems, cloud-integrated applications, and knowledge-driven system design.

WORK EXPERIENCE

Intern at Sony SSUP (AgroESP Project)

Dec 2024 - Sep 2025

- Developed backend-integrated IoT solutions for automated sensor onboarding and device management.
- Built a custom **OTA firmware update framework** for reliable microcontroller firmware delivery.
- Designed an **Automatic Sensor Registration (ASR) system** enabling plug-and-play deployment using ontology-driven discovery and LLM-assisted workflows.
- Worked with embedded systems, networking protocols, and automation pipelines for scalable IoT deployments.

Technical Intern and Web Developer, GalaxyZ Space (Startup)

Apr 2025 - Nov 2025

- Developed and deployed backend services using **Django** for educational workflows and platform management.
- Integrated **Supabase/PostgreSQL** for managing structured application data efficiently.
- Contributed to deployment, debugging, API integration, and maintenance of the organization's official website.

PROJECTS

Gyan – Hackathon Management Platform (Backend Developer)

- Engineered scalable backend services supporting event registration, team management, judging workflows, and project submissions.
- Designed and implemented secure **REST APIs** for frontend-backend integration.
- Worked on backend data handling, API routing, and workflow optimization for concurrent user operations.

Finova – Ontology-Driven Data Quality Risk Intelligence Platform

[GitHub](#)

- Developed a backend-driven data quality analysis platform for detecting completeness, uniqueness, and compliance issues.
- Implemented a reasoning engine to map data inconsistencies to business risk dimensions and generate actionable insights.
- Worked with structured knowledge representation, backend processing workflows, and scalable data handling concepts.

EDUCATION

2023 – 2027	B.Tech, Computer Science and Engineering Amrita Vishwa Vidyapeetham, Coimbatore	<i>CGPA: 9.04/10 (up to 6th Semester)</i>
2022 – 2023	Class 12th Board SRVS National Higher Secondary School, Karaikal, Puducherry	<i>Percentage: 95%</i>

SKILLS

Programming Languages	Python, C, C++, Java, JavaScript, Embedded C
Core CS Fundamentals	Data Structures & Algorithms, DBMS, Operating Systems, OOP, Computer Networks
Backend & Web Development	Django, REST APIs, HTML, CSS, JavaScript, Backend System Design
Databases & Data Management	PostgreSQL, SQL, Supabase, Knowledge Graphs, Semantic Data Organization
Systems & Networking	TCP/IP, HTTP/HTTPS, MQTT, Distributed Systems Fundamentals
Embedded Systems & IoT	OTA Updates, Sensor Integration, Arduino IDE, Keil µVision
Tools & Platforms	Git, Linux Fundamentals, Wireshark
Automation & Engineering Practices	Python Scripting, API Integration, Version Control, Debugging